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Sensors and Actuators

Laboratory Report
LED and Photodiode

Group 2 — Wednesday 14:00:

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1 Introduction

Understanding how light sensors work is crucial for explaining many everyday technologies. These sensors are used in devices such as smartphones, where they automatically adjust screen brightness based on ambient light conditions. They are also found in street lighting systems that turn on at night, in automatic headlights in cars, in cameras for exposure control, and in smart home systems that optimize energy usage by adapting lighting levels. Studying light sensors helps us understand how electronic systems interact with their environment and improves both efficiency and user experience.

The specific objective of this laboratory work is to use a red LED as an emitter and a photoresistor module as a receiver to detect when an object (e.g. a person) crosses the optical path between them, simulating a door-threshold detector. To this end, the LED is driven by one Arduino Uno while the photoresistor module is read by a second Arduino Uno; when the beam is interrupted, the analogue voltage at the module output changes abruptly and the event can be flagged.

To study this phenomenon, a **LED** and a **photoresistor module**, together with **two Arduino Uno microcontrollers** and a current-limiting **resistor**, were used.



Figure 1: Red 5 mm LED (left) and photoresistor module based on an LDR and an LM393 comparator (right), used respectively as emitter and receiver throughout this work.

2 Light-Emitting Diode (LED)

A LED (Light-Emitting Diode) is a semiconductive P–N junction that emits light when energized. That light is not monochromatic, as in a laser, instead, its spectrum occupies a narrow bandwidth. The light emitted by a LED is produced by the energetic interaction of the electrons in the material. In any directly polarized P–N junction, holes and electrons recombine. That recombination leads to a release of energy, as light or heat, by the free electron when it binds with the material. The wavelength of the emitted light depends on the band gap of the semiconductor material. By directly polarizing the junction one “pushes” the electrons and holes together, leading to their recombination.

The red LED used in the laboratory has the characteristics listed in Tab. 1.

Table 1: Specifications of the red 5 mm LED used as the emitter.

Parameter	**Value**
Diameter	5 mm
Viewing angle	$35°$
V_{nom}	$[1.8\,;\,2.2]$ V
I_{nom}	20 mA
I_{max}	50 mA
Luminosity (red)	$6300 - 8200$ mcd

3 The Photoresistor Module

A photoresistor (also known as a **Light-Dependent Resistor**, or **LDR**) is a passive electronic component whose **electrical resistance decreases as the intensity of incident light increases**. It is made of a high-resistance semiconductor material, typically cadmium sulfide (CdS), whose conductivity increases when photons are absorbed. In darkness the resistance can be several megaohms (MΩ), while in bright light it can fall to a few hundred ohms (Ω). The relationship between resistance and illuminance is approximately logarithmic and is described by the device's characteristic curve in the datasheet.

The **photoresistor module** used in this lab integrates an **LDR** with a **comparator circuit (LM393)** and an **adjustable threshold potentiometer**. It provides both a digital output (DO), which goes HIGH or LOW depending on whether the light level is above or below the set threshold, and an analogue output (AO) whose voltage depends on the LDR resistance, and hence on the ambient illuminance. The module operates from 3.3 V to 5 V and has four pins (VCC, GND, DO and AO), whose functions are:

- **VCC**: positive supply input, between 3.3 V and 5 V, used to power both the LDR voltage divider and the LM393 comparator.
- **GND**: ground reference, common to the supply and to the Arduino.
- **DO** (Digital Output): binary output of the LM393 comparator. It switches state when the light level crosses the threshold set by the on-board potentiometer.
- **AO** (Analogue Output): output of the voltage divider formed by the LDR and a fixed resistor on the module. Its voltage varies continuously with the incident illuminance and is the signal read by the Arduino's ADC in this work.

The complete set of electrical specifications is summarised in Tab. 2.

Table 2: Specifications of the photoresistor sensor module used as the receiver.

Parameter	**Value**
Material	Polymer
V_{nom}	$[3.3\,;\,5]$ V
Output type	Analogue and digital
Sensor type	Photoresistor (LDR)
Comparator	LM393
Adjustable threshold	Yes (via potentiometer)
Operating current	≈ 15 mA
Pins	VCC, GND, AO, DO

It is important to distinguish between nominal and operating values:

- **Operating value**: the actual value the device typically draws during normal use. It can vary depending on conditions (voltage, light level, circuit setup, etc.) and describes the typical consumption.
- **Nominal value**: a recommended, standard or rated value under standard conditions (a reference value defined by the manufacturer).

4 Circuit Assembly

The complete circuit used to drive the LED from the Arduino is shown in Fig. 2. It consists of the Arduino Uno 5 V supply, a current-limiting series resistor and the red LED. A second Arduino, not shown in this schematic, is used in parallel to read the analogue output of the photoresistor module (see Section 5).

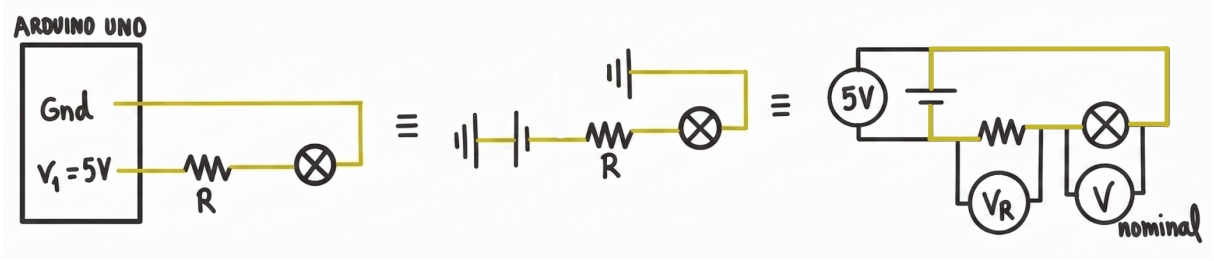


Figure 2: Electrical schematic of the LED driver circuit: the Arduino Uno digital pin supplies 5 V, a series resistor R limits the current, and the red LED is the load.

A series resistor is essential to limit the current flowing through the LED; without it, the LED could be damaged due to excessive current. The required resistance is obtained from Kirchhoff's voltage law applied to the loop,

$$V_{\text{Arduino}} = V_R + V_{\text{nom}}, \tag{1}$$

together with Ohm's law for the resistor,

$$V_R = R\, I_{\text{nom}}, \tag{2}$$

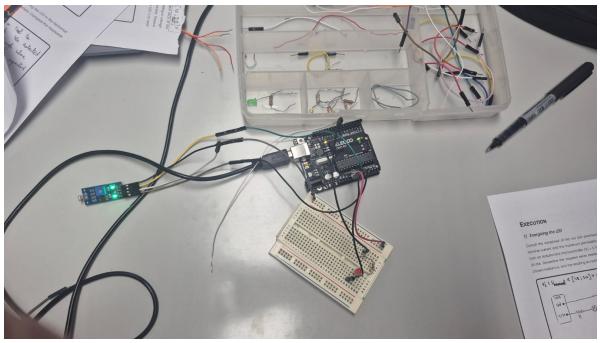
which, with $V_{\text{Arduino}} = 5$ V, yields

$$R = \frac{V_{\text{Arduino}} - V_{\text{nom}}}{I_{\text{nom}}}. \tag{3}$$

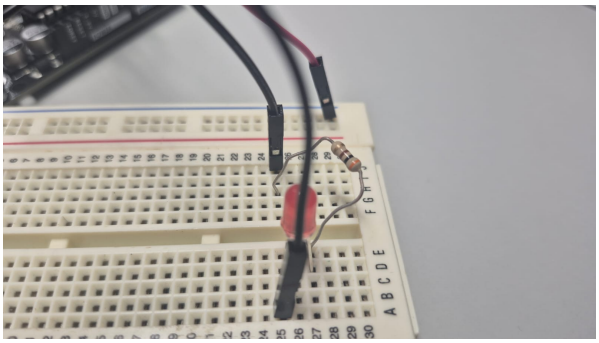
A nominal voltage of $V_{\text{nom}} = 2$ V was assumed, which lies within the normal operating range of the LED. With $I_{\text{nom}} = 20$ mA, equation (3) gives $R = 150\ \Omega$. Since a resistor with the exact calculated value was not available in the laboratory, the closest standard value greater than $150\ \Omega$ was selected: $R = 300\ \Omega$. Choosing a much higher resistance would significantly reduce the current supplied to the LED, resulting in lower brightness; choosing a lower one would damage the LED due to excessive current. With $R = 300\ \Omega$ the actual current through the LED becomes

$$I = \frac{V_{\text{Arduino}} - V_{\text{nom}}}{R} = \frac{5 - 2}{300} = 10\ \text{mA}, \tag{4}$$

which is half the nominal value but well below the maximum rating of 50 mA. This trade-off is acceptable: the LED is dimmer than at the nominal operating point, but the lifetime is extended and the rating is respected. With the circuit implemented, experimental validation can be performed in real conditions.



(a) Complete circuit with the Arduino connected to the host PC via USB.



(b) Close-up of the breadboard showing the series resistor and the LED.

Figure 3: Laboratory assembly of the LED driver circuit used in this work.

With the circuit assembled, multimeter measurements confirmed that the voltage across each component and the current through the loop matched the values computed above. A first sketch was then written in C++ to verify that the digital pin of the Arduino could drive the LED, by making it blink with a period of three seconds. The sketch is listed below; the variable `Serial` is used to print status messages on the host PC's serial monitor, while `digitalWrite` sets pin 8 to logic HIGH or LOW. The function `setup()` runs once at power-up and configures pin 8 as an output, and the function `loop()` is then executed indefinitely.

```
void setup() {
  Serial.begin(9600);          // initialise serial communication at 9600 baud
  pinMode(8, OUTPUT);          // configure digital pin 8 as an output
}

void loop() {
  digitalWrite(8, HIGH);       // turn the LED on
  Serial.println("light on");  // log the event on the serial monitor
  delay(3000);                 // wait 3 seconds

```

```
  digitalWrite(8, LOW);         // turn the LED off
  Serial.println("light off"); // log the event on the serial monitor
  delay(3000);                  // wait 3 seconds
}
```

Listing 1: Arduino sketch that makes the LED on pin 8 blink with a period of three seconds.

5 Analogue Reading of the Photoresistor Module

With the LED driver operating correctly, the photoresistor module readings were tested next. The module was placed at a fixed position facing the LED, and its analogue output AO was connected to the analogue input A0 of the second Arduino, as shown in Fig. 4.

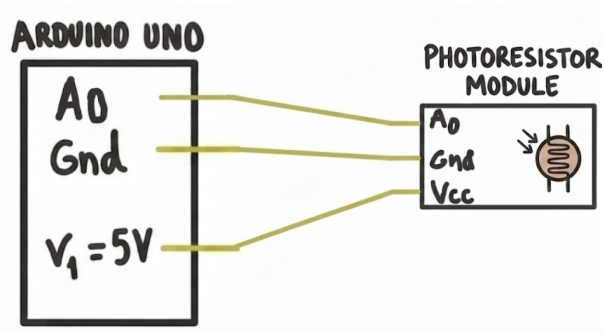


Figure 4: Connection of the photoresistor module to the second Arduino Uno: VCC and GND are taken from the Arduino's 5 V rail and the analogue output AO is read on pin A0.

The sketch listed below reads the ADC value on pin A0 every 500 ms and prints the corresponding voltage on the serial monitor. The variable `sensorPin` stores the analogue input pin number, while `lightValue` stores the 10-bit ADC reading (an integer between 0 and 1023). The conversion to volts is done by multiplying the raw reading by $5/1024$, the resolution of the Arduino's ADC.

```
int sensorPin = A0;                     // analogue input connected to AO
int lightValue = 0;                     // raw ADC reading (0--1023)

void setup() {
  Serial.begin(9600);                   // initialise serial communication
}

void loop() {
  lightValue = analogRead(sensorPin);         // read the LDR voltage divider
  Serial.print("Light Level (in volts): ");
  Serial.println(lightValue * 5.0 / 1024);    // convert ADC counts to volts
  delay(500);                           // wait half a second
}
```

Listing 2: Arduino sketch that reads the photoresistor module's analogue output and prints the corresponding voltage every 500 ms.

On the module, the LDR and a fixed resistor R_{fix} form a voltage divider supplied by $V_{\text{cc}} = 5\ \text{V}$, with the AO pin tapped across the LDR. The voltage seen by the Arduino is therefore

$$V_{\text{meas}} = V_{\text{cc}} \frac{R_{\text{var}}}{R_{\text{var}} + R_{\text{fix}}}, \tag{5}$$

where R_{var} is the LDR resistance and $R_{\text{fix}} \approx 10\ \text{k}\Omega$ is the fixed resistor on the module. Since the LDR resistance decreases as illuminance increases, V_{meas} is high in darkness and low under bright light.

Theoretical values. Typical datasheet values for CdS photoresistors of this type are $R_{\text{var}} \approx 1\ \text{M}\Omega$ in complete darkness and $R_{\text{var}} \approx 10\ \text{k}\Omega$ under moderate illumination such as the LED at 20 cm. Substituting in equation (5) gives

$$V_{\text{dark}} = 5 \cdot \frac{10^6}{10^6 + 10^4} \approx 4.95\ \text{V}, \tag{6}$$

$$V_{\text{light}} = 5 \cdot \frac{10^4}{10^4 + 10^4} = 2.50\ \text{V}. \tag{7}$$

Experimental values. In darkness the sensor measured $V_{\text{meas}} = 4.8\ \text{V}$, and with the LED at 20 cm $V_{\text{meas}} = 2.7\ \text{V}$. Both values lie within a few percent of the theoretical predictions, which validates the assumed divider configuration and the typical LDR resistance range.

6 LED Directivity

The LED datasheet specifies a viewing angle of $35°$, which corresponds to the full angle of the emission cone within which the radiant intensity remains above 50 % of its on-axis value. Equivalently, the half-intensity angle (the angular displacement from the optical axis at which the intensity has dropped to 50 % of the peak) is $\theta_{1/2} \approx 17.5°$.

To investigate the radiation pattern experimentally, the LED was placed facing the photoresistor module at a fixed distance, and a protractor was used to measure its angular orientation in the horizontal plane relative to the line joining the two devices. Starting from the on-axis position ($\theta = 0°$), the LED was slowly rotated while the analogue output V_{meas} was monitored on the serial console. The angle at which V_{meas} returned to the dark value $V_{\text{dark}} \approx 4.8\ \text{V}$ (which is the angle beyond which the LED no longer contributes appreciably to the illumination of the LDR) was found to lie between $40°$ and $50°$.

It is important to note that the quantity measured in this way is not the half-intensity angle $\theta_{1/2}$ but rather the *cut-off* (or extinction) angle θ_{cut}, i.e. the angle at which the residual contribution of the LED becomes indistinguishable from the ambient illumination level. For a typical 5 mm LED with a roughly Gaussian or Lambertian-shaped radiation pattern, the cut-off angle is expected to be approximately twice the half-intensity angle, around $2\theta_{1/2} \approx 35°$, or a little more once the finite sensitivity of the LDR to weak residual light is taken into account.

The relatively coarse precision of the measurement is due to:

- (i) the gradual roll-off of the radiation pattern near its edge, which makes the transition to the ambient value difficult to pinpoint with a protractor;
- (ii) the finite active area of the LDR, which averages the intensity over a small solid angle;

- (iii) stray reflections from the bench and surrounding objects.

A finer characterization would require sampling V_{meas} at small angular steps over the full range and plotting the normalized intensity $I_n(\theta) = [V_{\text{dark}} - V_{\text{meas}}(\theta)]/[V_{\text{dark}} - V_{\text{meas}}(0)]$, from which $\theta_{1/2}$ could be read directly at the $I_n = 0.5$ level.

7 Application: Door-Threshold Detector

With the optical link characterised, the system was configured as a simple presence detector. When an opaque object passes between the LED and the photoresistor module, the beam is blocked and V_{meas} rises sharply towards V_{dark}. A threshold voltage V_{th} was chosen midway between the illuminated and dark levels,

$$V_{\text{th}} = \frac{V_{\text{light}} + V_{\text{dark}}}{2} \approx 3.75\ \text{V}, \tag{8}$$

and the second Arduino flags an event whenever $V_{\text{meas}} > V_{\text{th}}$. A simple state variable, `blocked`, is used to detect only the rising edges (i.e. the instants in which the beam is interrupted) and avoid printing the message continuously while the beam is blocked. The corresponding sketch is shown below.

```
int sensorPin = A0;            // analogue input connected to A0
const float V_TH = 3.75;       // threshold voltage in volts
bool blocked = false;          // current state of the beam

void setup() {
  Serial.begin(9600);
}

void loop() {
  float v = analogRead(sensorPin) * 5.0 / 1024.0;  // ADC reading converted to
    volts

  if (v > V_TH && !blocked) {                      // rising edge: beam
    interrupted
    Serial.println("Crossing detected");
    blocked = true;
  } else if (v < V_TH && blocked) {                // falling edge: beam restored
    blocked = false;
  }

  delay(50);                                       // 20 Hz sampling rate
}
```

Listing 3: Arduino sketch implementing the door-threshold detector using edge detection on the photoresistor analogue output.

8 Conclusion

In this work a red LED was driven from an Arduino Uno through a current-limiting resistor and used to illuminate a photoresistor module read by a second Arduino. The series resistor

was sized to keep the forward current at a safe 10 mA, half of the nominal value but well below the absolute maximum rating, prioritizing LED longevity over peak brightness. The analogue output of the module was shown to follow the expected voltage-divider behavior, with measured values agreeing with theory to within a few percent. The LED radiation pattern was characterized by rotating the emitter about the receiver with a protractor, yielding a cut-off angle of $[40°,50°]$, which is consistent with the $35°$ viewing angle stated in the datasheet once the difference between cut-off and half-intensity angle is taken into account, since the cut-off angle is larger than the half-intensity angle. Finally, the system was used to implement a simple door-threshold detector, demonstrating that a LED–LDR pair is sufficient for low-cost presence sensing.

References

[1] F. Alegria, *Sensors and Actuators*, sections 8.4 and 8.9.

[2] Bot 'n Roll, "5 mm red LED, 6000 mcd," product page. `https://www.botnroll.com/pt/5mm/256-497-led-azul-5mm-alto-brilho-6000mcd.html`.

[3] Keszoox, "Photosensitive Sensor Module (LM393)," product page. https://keszoox.com.

[4] Texas Instruments, "LM393 Dual Differential Comparator," datasheet, 2017. `https://www.ti.com/product/LM393`.

[5] Arduino, "Arduino Uno Rev3," product page. `https://store.arduino.cc/products/arduino-uno-rev3`.

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